

Kevin Huang Wu

Technology & Interactive Systems Graduate

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kewusir.github.io/KewusirPortfolio/ | Irish citizen - unrestricted right to work in Ireland and the EU

Profile

Technology and interactive-systems postgraduate at Trinity College Dublin with a technical progression from Music Technology through Music and Media Technologies to Interactive Digital Media. Hands-on experience across C#, Python, JavaScript, front-end development, Git, Unity, real-time data processing, Arduino, EMG/EEG integration, user research and testing. Builds end-to-end prototypes spanning hardware assembly, software integration, debugging, user evaluation and data analysis. Uses AI- and MCP-enabled workflows as practical development aids while independently reviewing, testing and validating outputs. Native bilingual proficiency in English and Mandarin Chinese.

Technical Skills

Programming & Web: C#, Python, JavaScript, HTML5, CSS3, jQuery, Git/version control, JSON, front-end development.

Software & Interactive Systems: Unity, real-time interaction, state/input logic, debugging, functional testing, rapid prototyping, VR/PCVR, HCI.

AI-Enabled Development: Generative AI, AI-assisted coding and debugging, MCP-enabled workflows, prompt-based technical problem solving, output validation, critical evaluation of AI-generated text, imagery and feedback.

Hardware & Physical Computing: Arduino Uno R3, EMG sensing, OpenBCI/EEG, sensor integration, soldering, analogue data acquisition, calibration, hardware-software troubleshooting.

Research & Analysis: User research, user testing, requirements investigation, literature/contextual research, REAMS/ethics documentation, experimental design, qualitative and quantitative data analysis.

Creative Technology: Pure Data (Pd), Wwise, Logic Pro, spatial/binaural and 5.1/7.1 audio, Adobe Creative Cloud, Blender, semiotics, branding and visual communication.

Selected Technical & Research Experience

Technical Developer & Project Manager - Biometric PCVR Horror Experience

MSc Interactive Digital Media team project, Trinity College Dublin | 2026

- Built a custom EMG-based interaction system by assembling and soldering the sensor/Click-board hardware, integrating it with an Arduino Uno R3, and connecting real-time physiological input to Unity/C# interaction logic.
- Iteratively calibrated and debugged analogue EMG readings, implementing threshold detection, re-arming/state logic and fallback controls while investigating false triggers, missed inputs and reliability edge cases.
- Led technical delivery within a five-person multidisciplinary team, coordinating implementation across Unity, hardware, narrative, 3D assets, Wwise/spatial audio, testing and final showcase preparation.
- Used AI-assisted and MCP-enabled development workflows to accelerate prototyping, code iteration and troubleshooting while manually validating generated code, system behaviour and hardware-software integration.

Researcher & Unity Developer - Immersive Audio User Study

MSc Interactive Digital Media research, Trinity College Dublin | 2026

- Conducted literature, background and contextual research to define research questions, user requirements and an experimental approach for evaluating spatial-audio formats and immersion.
- Prepared research ethics documentation through Trinity's REAMS process and developed participant-facing study materials, procedures and evaluation measures.
- Built the Unity-based test environment, conducted user testing, and collected and analysed qualitative and quantitative data to evaluate localisation, immersion and user experience.

Unity Developer & HCI Researcher - The Spirit Land

MPhil dissertation project, Trinity College Dublin | 2025

- Designed and developed a first-person Unity application using OpenBCI EEG signals to drive adaptive interaction through attention and relaxation metrics.
- Built a real-time Python-to-Unity data pipeline and implemented configurable thresholds, UI/audio feedback and manual fallback controls to maintain usability under variable biometric input.
- Planned and conducted participant evaluation, analysing usability, interaction reliability and immersion to inform iterative design improvements.

Leadership & Commercial Experience

Chairperson - Trinity College Dublin Chinese Society

Trinity College Dublin | 2023-Present

- Lead a society of 500+ members and a social-media audience of 1,300+, overseeing committee operations, planning, communications, member engagement and external stakeholder relationships.

- Built and managed partnerships with 20+ Dublin businesses across hospitality, retail, personal services and food-delivery sectors, securing member offers while providing partners with targeted exposure to the Chinese student community.
- Coordinated a member engagement with the Montblanc boutique in Dublin, connecting the brand with Chinese international students through an organised in-store experience and preferential offering.
- Led end-to-end delivery of a 250-attendee formal event at Clontarf Castle Hotel with a €14,000+ budget, managing venue liaison, suppliers, ticketing, promotion, team delegation, logistics and on-site operations.

Education

MSc Interactive Digital Media

Trinity College Dublin | 2025-2026

Technical focus: JavaScript, HTML/CSS, front-end development, Git, interactive programming, computer graphics, Unity/C#, XR, sensor technologies and HCI. Additional study included user research, data analysis, semiotics and branding, and critical analysis of generative-AI outputs and feedback.

MPhil Music and Media Technologies - Merit

Trinity College Dublin | 2024-2025

Technical focus: Unity/XR, Python-enabled real-time data workflows, interactive media, psychoacoustics, audio engineering, spatial/multichannel audio and computational approaches to audiovisual systems.

BA Music - Second Class Honours, First Division (2.1)

Trinity College Dublin | 2020-2024

Music Technology specialisation with technical study in digital audio, synthesis/sampling, sound design, audiovisual production and patch-based programming in Pure Data (Pd).

Languages

English: Native bilingual proficiency

Mandarin Chinese: Native bilingual proficiency